PAWS Installation and Setup Guide

Overview

PAWS (Physical Altercation and Weapon Surveillance System) is an automated surveillance application designed to detect physical altercations and weapons in video feeds using computer vision and deep learning models.

The system supports:

- Physical altercation detection using VideoMAE
- Weapon detection for knives and guns
- Real time event logging
- Automated evidence capture
- GPU accelerated processing with CUDA support

System Requirements

Recommended Hardware:

Component	Specification
CPU	Intel Core i5 or higher
RAM	16GB or higher
Storage	256GB or higher
GPU	NVIDIA RTX 3050 or higher

Supported Software

- Windows 10 or Windows 11
- Python 3.9 or later
- CUDA 11.8 or later (recommended for NVIDIA GPUs)

Required Models

Download the required model files from the provided Google Drive link.

Place the files inside the models/ folder.

Required structure:

```
models/
├── videomae-violence-local/
│   ├── config.json
│   ├── model.safetensors
│   └── preprocessor_config.json
├── yolov8s.pt
├── knife_detector_adamw.pt
├── mobilenet_violence_prefilter.pt
└── cctv_gun_detector.pt
```

Installation Guide

1. Clone the Repository

git clone https://github.com/YanKylee/PAWS-Physical-Altercation-Weapon-Surveillance-System.git

cd PAWS-Physical-Altercation-Weapon-Surveillance-System

2. Create and Activate a Virtual Environment

python -m venv venv

venv\Scripts\activate

3. Install PyTorch

GPU Version (CUDA 11.8 Example)

pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118

CPU Only Version

pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu

4. Install Required Packages

pip install ultralytics opencv-python customtkinter pillow pygame transformers

Verification

Check if PyTorch detects CUDA correctly:

python -c "import torch; print(torch.__version__, torch.version.cuda, torch.cuda.is_available())"

Check Transformers installation:

python -c "import transformers; print(transformers.__version__)"

Running the Application

Run the application using:

python main_ui.py

Troubleshooting (CUDA Not Detected)

If torch.cuda.is_available() returns False:

- Confirm NVIDIA drivers are installed
- Confirm CUDA is installed
- Ensure the installed PyTorch version matches the CUDA version

Missing Model Files

Verify all required model files are correctly placed inside the models/ directory.

Missing Python Packages

Reinstall dependencies:

pip install ultralytics opencv-python customtkinter pillow pygame transformers

Slow Performance

- GPU acceleration is strongly recommended
- CPU only mode will process videos significantly slower

Notes

- The application uses local VideoMAE model files through the transformers library.
- Internet is not required if all models are stored locally.
- Detection accuracy depends on lighting conditions, video quality, and camera positioning.